

Do Brain-Inspired Mechanisms Help Continual Reinforcement Learning?

A Controlled Ablation Study with Spiking Agents

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Abstract

Brain-inspired mechanisms such as spiking neurons, local plasticity, homeostasis and sleep-like consolidation are proposed as remedies for catastrophic forgetting in continual reinforcement learning (RL), usually several at once. We ask which of them help when each is isolated and compared with a matched non-biological control. In task-incremental MiniGrid sequences with small PPO agents, we test seven mechanisms one at a time, with pre-registered tests (3–10 seeds per arm, 95% confidence intervals) and matched budgets. The answer is mostly no. Reward-modulated STDP is indistinguishable from a random update of the same size (AUC difference $+0.033$ $[-0.260, +0.326]$). Sleep-like consolidation removes most of naive fine-tuning’s forgetting but, at an exactly matched replay budget, ties experience replay (final-accuracy difference $+0.017$ $[-0.026, +0.060]$). There are two specific exceptions. Homeostasis speeds up learning in a spiking encoder trained without backpropagation, by direct feedback alignment (AUC difference $+0.326$ $[+0.101, +0.551]$, Holm $p = 0.017$), but does not make it reliable. With a small replay buffer, a shared network with sleep reaches similar mean accuracy to one network per task (0.838 vs 0.833; 3 seeds, wide CI) in less than half the total memory; at the default buffer, isolation is more memory-efficient. A spiking encoder works at fewer operations per frame than a sparsified CNN, but costs $5.2\times$ more CPU time to train; its energy advantage is an estimate. Whether Transformer working memory helps remains untested: no benchmark we tried both required memory and was learnable within budget. Six earlier positive findings shrank or disappeared under controls; we report each with its cause.

1 Introduction

A continual reinforcement learning (RL) agent learns a sequence of tasks while its parameters keep changing. With plain gradient-based fine-tuning, learning a new task overwrites what earlier tasks needed: catastrophic forgetting. The problem is framed as a stability–plasticity trade-off [LIT] (Khetarpal et al., 2022; Kirkpatrick et al., 2017), and the standard remedies are regularisation towards old parameters (EWC; Kirkpatrick et al., 2017), replaying stored experience (CLEAR; Rolnick et al., 2019) and isolating parameters per task (e.g. PackNet; Mallya & Lazebnik, 2018).

A second family of remedies is inspired by the brain, and it is popular. Spiking neural networks (SNNs) promise event-driven, low-energy computation [LIT] (Fang et al., 2023; Zhou et al., 2026). Local learning rules such as STDP and its reward-modulated three-factor variants are proposed as biologically plausible complements or alternatives to backpropagation [LIT] (Lee et al., 2025; Huo et al., 2025). Sleep-like offline replay with local plasticity has been reported to recover forgotten tasks [LIT] (Tadros et al., 2022), and homeostatic plasticity keeps biological firing rates in a working range [LIT] (Turrigiano & Nelson, 2004). These mechanisms are often introduced together, in one system, and compared against an unmodified baseline. When such a system does better, it is hard to know which mechanism did the work, or whether a simpler non-biological control would have done the same.

We ask a narrower question: *in a controlled continual RL setting, which brain-inspired mechanisms improve learning or retention when each is isolated and compared with a matched control?* We build one agent (an SNN or CNN encoder, optional Transformer working memory, PPO) and switch seven mechanisms on and off: the spiking encoder, STDP, homeostasis, direct feedback alignment (DFA) as a global learning signal that does not need backpropagation’s weight transport, sleep-like consolidation, shared weights versus parameter isolation, and Transformer working memory. Each gets a control chosen to remove the most obvious alternative explanation: a same-size random update for STDP, a replay baseline with an exactly matched number of replayed samples for sleep, total-memory accounting for shared weights, a memoryless baseline and a frame stack for the Transformer.

Headline answer. [OURS] Mostly no, with two specific exceptions. No mechanism improved final accuracy or forgetting beyond its matched non-biological control. The exceptions are narrower than that: homeostasis speeds up learning when the spiking encoder is trained by DFA instead of backpropagation, without making it reliable; and with a small replay buffer, a shared network with sleep reaches similar mean accuracy to one network per task (0.838 vs 0.833; 3 seeds, wide CI) in less than half the memory; at the default buffer size, isolation is more memory-efficient. The spiking encoder reaches a lower operation-count regime than a sparsified CNN, at a small accuracy cost; whether that saves energy depends on neuromorphic hardware we did not use. The Transformer question remains untested, because no benchmark we tried both required memory and was learnable within budget.

Contributions.

1. A controlled ablation of seven brain-inspired mechanisms for task-incremental continual RL, each switched on alone and compared with a control chosen to rule out its most obvious alternative explanation (Sections 4 and 5).
2. Two budget-matched comparisons that change the conclusion of the unmatched ones: sleep versus replay at an exactly matched number of replayed samples, and shared weights versus isolation per unit of total memory (parameters plus replay buffer) rather than per parameter (Sections 5.4 and 5.5).
3. Pre-registered tests with effect sizes, 95% confidence intervals and multiple-comparison correction (Section 3.4), and a record of six earlier positive findings that shrank or disappeared under controls, each with its cause (Section 6).
4. Code, frozen results, a script that regenerates every number in this paper from the result files, and an offline results page with the trained agents (Reproducibility statement).

How to read the claims. We label statements as literature claims [LIT], results of our experiments [OURS], or our interpretations [INTERP]. Literature numbers were taken from the cited papers and are marked as such.

2 Related work

Continual RL. [LIT] Khetarpal et al. (2022) review continual RL and its desiderata: retention, forward transfer, and ideally operation without task identities or boundaries. EWC (Kirkpatrick et al., 2017) penalises moving parameters that were important for earlier tasks (Fisher-weighted); it protects old tasks at the cost of plasticity for new ones. CLEAR (Rolnick et al., 2019) mixes new experience (for plasticity) with replayed old experience (for stability), trains on both with the off-policy V-trace actor-critic loss, and adds behavioural cloning on the replayed experience: a KL term towards the stored policy and an L2 term towards the stored value. It needs no task identities, and it is a strong, simple continual RL method. Our replay baseline and our sleep phase both reuse CLEAR’s cloning losses; Section 4.5 states exactly how they differ from CLEAR and from each other. Continual World (Wolczyk et al., 2021) argues that forgetting alone is an incomplete metric and that forward transfer must be measured; we report their forward-transfer measure. PackNet

(Mallya & Lazebnik, 2018) isolates parameters per task by pruning and freezing, which removes forgetting by construction; our isolation baseline is the simpler one-network-per-task variant. We measure forgetting as in Chaudhry et al. (2018).

Sleep and consolidation. [LIT] Tadros et al. (2022) implement a sleep-like phase in feed-forward networks (offline, noisy input, local unsupervised Hebbian plasticity) and report that it recovers previously forgotten tasks in incremental classification. Our sleep phase is *inspired* by this line of work but is different: it is offline replay plus distillation in an RL setting, optionally with STDP (Section 4.5), and it relies on a replay buffer. Yu et al. (2025) distil accumulated experience of LLM agents into parameters instead of retrieving raw trajectories, which motivates counting the memory that replay needs (Section 5.5).

Spiking networks for RL and control. [LIT] SNNs are trained end to end with surrogate gradients (Neftci et al., 2019). Van den Berghe et al. (2025) study the surrogate-gradient slope and show that it matters much more in RL than in supervised learning; they also describe a warm-up problem for stateful SNNs in RL. Our encoder resets its membrane at every environment step, so it avoids that problem, and it has no memory across frames (Section 4.1). Lee et al. (2025) and Huo et al. (2025) review biologically plausible learning rules for SNNs, including STDP and reward-modulated three-factor rules (Bi & Poo, 1998; Frémaux & Gerstner, 2016). The Spiking Decision Transformer (Pandey & Biswas, 2025) combines spiking attention with three-factor plasticity for offline RL on classic control tasks; Spikingformer (Zhou et al., 2026) points out that many “spiking” architectures hide non-spike floating-point computation, a warning we take into account when counting operations. SpikingJelly (Fang et al., 2023) and snnTorch (Eshraghian et al., 2023) are common SNN toolkits; we use our own LIF implementation, checked against snnTorch.

Local and weight-transport-free learning. [LIT] Direct feedback alignment (Nøkland, 2016) projects the output error to hidden layers through fixed random matrices, avoiding the weight transport that back-propagation requires. Homeostatic plasticity (Turrigiano & Nelson, 2004) adjusts excitability to keep firing rates near a set point.

Energy proxies. [LIT] A common proxy counts synaptic operations and charges an accumulate (about 0.9 pJ) per spike-driven operation and a multiply-accumulate (about 4.6 pJ) per dense operation, using 45 nm estimates (Horowitz, 2014). Such numbers are estimates, not measured power, and they assume event-driven hardware.

3 Setup

3.1 Environments and task sequences

All environments are MiniGrid (Chevalier-Boisvert et al., 2023) gridworlds with the standard 7×7 egocentric symbolic view, which we encode as 20 one-hot channels (object type, colour and state).

fetch3 and fetch5 (continual). A room (6×6 interior) contains three (fetch3) or five (fetch5) distinct objects at random positions. Task k rewards picking up object k (reward $1 - 0.9t/T_{\max}$, the MiniGrid convention); picking up any other object ends the episode with reward 0. Observations are statistically identical across tasks, so the same state demands different actions in different tasks: the setting guarantees interference under naive fine-tuning, while navigation and pick-up are shared skills that allow forward transfer. The agent is given the task identity (a task embedding and one actor-critic head per task). This is the standard *task-incremental* setting; without the task identity the tasks are not merely harder but unsolvable, since the goal is not visible (Section 7). Tasks are trained in order for a fixed number of frames each (150k per task for the CNN and SNN agents, 400k for the SNN+Transformer agent).

DoorKey-6x6 (single task). Used for the local-plasticity questions: the agent must pick up a key, open a door and reach the goal; 400k frames. It is hard enough that the encoder’s learning matters (a frozen random encoder learns more slowly), and outcomes are bimodal (a run either solves it or stays near zero).

MiniGrid-Memory (S7, S11, S13). The agent sees a cue object at one end of a corridor and must choose the matching object at the other end. We use these maps for the Transformer question (Section 5.6).

3.2 Agent and training

The encoder is either a CNN (three 2×2 convolutions with 16, 32 and 64 channels and a linear layer to 128 features) or an SNN with the same topology, leaky integrate-and-fire (LIF) neurons, 4 timesteps per environment step, a learnable leak (initial $\beta = 0.9$) and a surrogate gradient (slope 25) (Neftci et al., 2019). Actor and critic heads are small MLPs. Training is PPO (Schulman et al., 2017) in a single-file CleanRL-style implementation (Huang et al., 2022): 16 parallel environments, 128 steps per rollout, 4 epochs, minibatches of 256, clip 0.2, $\gamma = 0.99$, GAE $\lambda = 0.95$, early stopping at an approximate KL of 0.03, Adam with learning rate 0.001 (0.0003 for the SNN+Transformer agent). Everything runs on CPU (one thread per run). The shared CNN agent has 194,312 parameters, one isolated CNN network 159,864, the shared SNN+Transformer agent 327,800.

3.3 Metrics

$R_{i,j}$ is the mean evaluation return on task j after training on task i (100 episodes, actions sampled from the policy). For T tasks: $\text{ACC} = \frac{1}{T} \sum_j R_{T,j}$ (final average performance); $\text{BWT} = \frac{1}{T-1} \sum_{j < T} (R_{T,j} - R_{j,j})$; $\text{FORGET} = \frac{1}{T-1} \sum_{j < T} (\max_{i < T} R_{i,j} - R_{T,j})$ (Chaudhry et al., 2018); forward transfer FWT is the normalised gain in area under the training curve on tasks $2..T$ relative to a from-scratch network on the same task and seed (Wolczyk et al., 2021). For single-task runs we report the solve rate (final evaluation return above 0.5 over 200 episodes) and the AUC (mean training return over the run, a sample-efficiency measure).

Memory accounting. For the shared-weights question we count total memory as fp32 parameters plus the replay buffer at its measured storage cost: 188 bytes per stored state for the CNN and SNN agents (observation, mask, task id, teacher logits and value) and 632 bytes for the SNN+Transformer agent (4-frame windows), times 5,000 states per task by default.

3.4 Statistics

Each arm has 3–10 seeds. For solve rates we use Fisher’s exact test and the Newcombe (Wilson score) interval for the difference of proportions (Newcombe, 1998); for continuous metrics, Welch’s t -test (Welch, 1947) and the Welch 95% confidence interval of the difference of means. Where a pre-registered family contains several contrasts we apply Holm’s correction (Holm, 1979). With so few seeds, CIs are wide; we treat them as the primary result and read a non-significant difference as “no large effect”, not as equivalence.

Pre-registration. The work was done in four phases. The first two were exploratory. For the last two, the comparisons, test statistics, seed counts and stopping rules were recorded in the project’s version-controlled log before the runs, one planned test per question. This is a project record, not a public registry. The tests we call *pre-registered* below are those. Exploratory runs added later are labelled as such and enter no test.

4 Mechanisms and their controls

Table 1 lists each mechanism, how it is implemented and what it is compared against.

4.1 Spiking encoder

The SNN integrates each frame over 4 internal timesteps and its membrane state is reset at every environment step, so it has no memory across frames; we verified this empirically (identical output for identical frames regardless of history). We count event-driven operations per frame: synaptic operations for the SNN (non-zero presynaptic spikes times fan-out, summed over timesteps) and non-zero-activation multiply-accumulates for a CNN whose activations are sparsified with an L1 penalty. The energy proxy multiplies these counts

Table 1: Mechanisms, implementations and controls. Each row is switched on and off independently.

Mechanism	Implementation	Control / comparator
Spiking encoder	LIF CNN, 4 timesteps per frame, surrogate gradients; membrane reset every environment step	Same-topology CNN; op count at matched accuracy
STDP	Reward-modulated three-factor STDP on encoder synapses, added to the backprop update: $\Delta w = \alpha \Delta w_{\text{STDP}} + \Delta w_{\text{grad}}$, stabilised by mean-centring and homeostasis	Same-RMS random update + homeostasis; homeostasis alone
Homeostasis	Per-neuron bias pushes firing rate towards 0.15	Same agent without it
DFA	Encoder layers learn from the output error through fixed random feedback (heads exact)	DFA without homeostasis; backprop
Sleep	Offline phases: old-task distillation on stored states + current-task self-distillation	CLEAR-style replay (cloning losses, interleaved), replayed samples matched exactly; EWC; naive
Shared weights	One trunk, per-task heads	One network per task (isolation), at equal total memory
Transformer memory	Transformer over the last K encoded frames (past frames only)	Memoryless agent; frame stack of K frames

by the literature constants above. To compare accuracy against operations without RL noise, the op-count frontier is measured by behaviour cloning of a trained DoorKey-6x6 teacher: both encoders plus an identical linear readout are trained with an activity penalty swept over a range, and we report held-out action accuracy and the closed-loop return of the cloned policy (2 seeds per point).

4.2 STDP and the random-update control

After every PPO optimiser step, spikes recorded in the minibatch’s forward pass produce a local STDP update for each encoder layer (eligibility traces, pre/post pairing), multiplied by a third factor, the normalised PPO advantage of the sample (reward-modulated STDP) (Frémaux & Gerstner, 2016; Lee et al., 2025). The update is normalised to unit RMS and scaled by α times the learning rate, so $\alpha = 1$ moves each weight about as much as one Adam step. The *stabilised* version mean-centres pre- and post-synaptic spikes (a covariance rule, no drift under independent firing) and adds homeostasis. The *random* control applies an update with the same per-layer RMS but a random direction, also with homeostasis, so it separates “STDP helps” from “any perturbation of this size helps”. We use $\alpha = 0.03$ for the pre-registered comparison.

4.3 Homeostasis

Each spiking neuron’s bias is nudged towards a target firing rate: $b \leftarrow b + \eta(0.15 - r)$ with $\eta = 0.01$ and r the neuron’s rate in the minibatch.

4.4 Direct feedback alignment

In the DFA agents, the encoder’s layers receive the output error through fixed random feedback matrices instead of backpropagated gradients (Nøkland, 2016); the actor and critic heads still use exact gradients. This gives a global learning signal without weight transport.

4.5 Sleep-like consolidation and the replay baseline

Every 25 PPO updates, and once at the end of each task, the agent goes offline for 700 gradient steps on two losses: (i) distillation towards the stored policy and value on states saved at the end of each earlier task (KL divergence plus value MSE; the same buffer the replay baseline uses), and (ii) self-distillation of the current task towards a frozen snapshot of the network taken at sleep onset, on a reservoir of recent wake states. Optionally, local STDP runs on the spikes produced during sleep. The *replay* baseline adds the same distillation loss on one replayed minibatch to every PPO minibatch while learning later tasks, as in CLEAR.

Relation to CLEAR. Both methods reuse CLEAR’s behavioural-cloning losses (Rolnick et al., 2019), and they differ from CLEAR, and from each other, in specific ways. (i) Our *replay* baseline keeps CLEAR’s cloning terms (policy KL and value regression on replayed states) but not its off-policy RL loss on replayed experience: PPO is on-policy, and replayed transitions cannot enter its objective without an importance-weighting scheme such as V-trace, which we did not add. The cloning targets are the policy and value of the network at the end of the task a state comes from, stored with the state, rather than those of the policy that generated the experience. As in CLEAR, replay is interleaved with every update on the new task. (ii) Our *sleep* phase uses the same buffer and the same cloning losses, but applies them offline, in scheduled phases without environment interaction, and adds self-distillation on recent current-task states. With the number of replayed states matched, the sleep–replay comparison therefore isolates *when* replay happens, plus the self-distillation term; it does not compare sleep with the full CLEAR method. (iii) Unlike CLEAR, both use the task identity (per-task heads), because our setting is task-incremental.

Replay budget. The two methods consume different numbers of replayed states. Replay’s count depends on how many PPO minibatches run, which varies by seed because of KL early stopping (0.91M–1.48M per SNN+Transformer run); sleep’s is fixed by its schedule (1.43M). For the pre-registered comparison, each sleep run was given a total replay budget equal to the measured count of the same-seed replay run, spread evenly over its sleep phases (*sleep, replay-matched*); the match was exact for every seed.

4.6 Shared weights versus isolation

The shared agent keeps one trunk and adds a head per task. Isolation trains a fresh network per task: zero forgetting by construction and parameters linear in the number of tasks. We compare them per parameter, as is common, and per unit of total memory, which includes the replay buffer that the shared methods need.

4.7 Transformer working memory

The SNN+Transformer agent encodes each of the last K frames and runs a one-layer Transformer encoder (Vaswani et al., 2017) over them, reading out the newest position; the window holds only past frames, and padding before the episode start is masked. On the fetch tasks $K = 4$. The non-attention baselines are a frame stack (the last K frames as input channels) and, implemented but not run, a GRU over the same encoded frames.

5 Results

5.1 Spiking encoder: a lower operation-count regime, not more accuracy

[OURS] Where both encoders work, they are equally accurate at matched operation counts (Figure 1). The unpenalised CNN uses 210k event-driven operations per frame (accuracy 0.999). The L1-sparsified CNN still works at 33k operations (accuracy 0.998), but a stronger penalty silences the whole network at once (6.9k operations, accuracy 0.524, the majority-action rate). The SNN keeps working below that floor: with 4 timesteps reduced to 2 it reaches 16.6k operations at accuracy 0.986 and a closed-loop return of 0.970, about $2.0\times$ fewer operations than the CNN floor; $T = 4$ behaves similarly (16.2k, 0.969). Under the literature energy constants this point costs 14.9 nJ per frame against 126 nJ for the sparsest working CNN and 1653 nJ for the dense CNN.

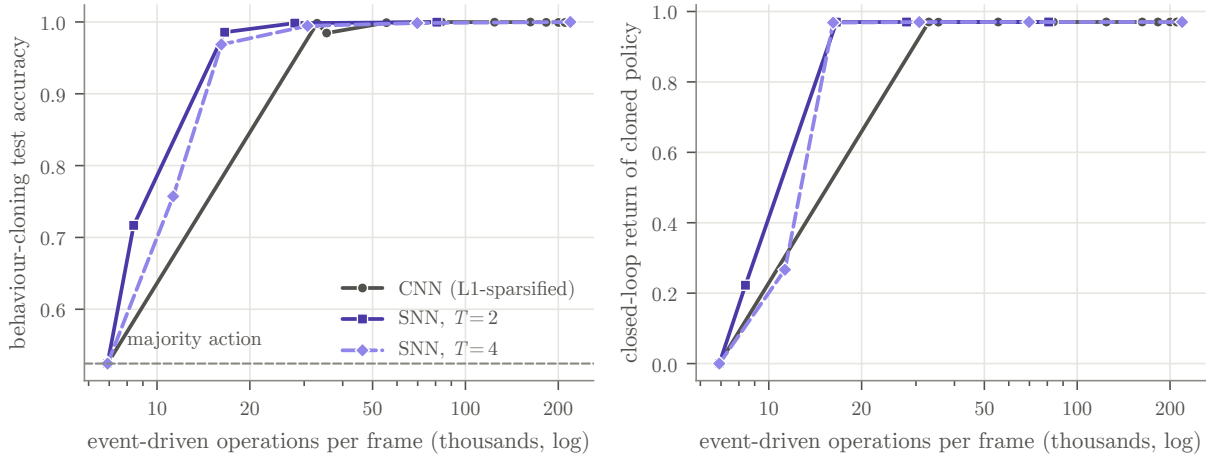


Figure 1: Accuracy against operation count for the spiking and the conventional encoder. Each encoder, with an identical linear readout, is trained by behaviour cloning of a trained DoorKey-6x6 PPO teacher with an activity penalty; each point is one penalty value (mean of 2 seeds). The x -axis counts event-driven operations per frame (synaptic operations for the SNN, non-zero multiply-accumulates for the L1-sparsified CNN; log scale). Left: held-out action accuracy (dashed: majority-action rate). Right: closed-loop return of the cloned policy. The SNN keeps working below the lowest operation count at which the CNN still works.

Table 2: Pre-registered local-plasticity contrasts on DoorKey-6x6 (10 seeds per arm). Differences are the first arm minus the second, with 95% CIs (Newcombe for solve rates, Welch for AUC). Holm-corrected p across the four contrasts, separately per statistic.

Contrast	Δ solve rate [95% CI]	Fisher p (Holm)	Δ AUC [95% CI]	Welch p (Holm)
STDP vs homeostasis alone	-0.20 [-0.51, +0.11]	0.474 (1.000)	-0.101 [-0.323, +0.121]	0.338 (0.724)
STDP vs matched random	+0.00 [-0.34, +0.34]	1.000 (1.000)	+0.033 [-0.260, +0.326]	0.818 (0.818)
homeostasis vs backprop	+0.20 [-0.11, +0.51]	0.474 (1.000)	+0.200 [+0.024, +0.377]	0.029 (0.117)
random vs homeostasis	-0.20 [-0.51, +0.11]	0.474 (1.000)	-0.134 [-0.372, +0.104]	0.241 (0.724)

[INTERP] The SNN’s advantage is access to a lower-operation regime, not better accuracy, and the energy numbers assume event-driven hardware. Other CNN sparsifiers (e.g. k -winner-take-all) might move the CNN floor; we did not test them. On the CPU we actually used, the SNN costs 80 ms per training step against 16 ms for the CNN ($5.2\times$), because it simulates 4 timesteps.

Robustness per unit of compute. [OURS] Dividing final continual accuracy (fetch3) by training compute (per-step CPU cost times frames), the plain CNN scores 0.978 per unit, $5.7\times$ the best spiking variant (SNN with sleep, 0.173); SNN with homeostasis scores 0.164, DFA with homeostasis 0.090, stabilised STDP 0.084 and the SNN+Transformer agent 0.056. Encoders trained by local STDP only never solved DoorKey (0/6 runs).

5.2 STDP: no effect beyond a random update of the same size

[OURS] With backpropagation as the global signal, the four arms solve DoorKey-6x6 at similar rates: backprop 8/10, backprop with homeostasis 10/10, with stabilised STDP 8/10, with the matched random update 8/10 (Figure 2). All four pre-registered contrasts are null after Holm correction (Table 2). STDP is indistinguishable from a random update of the same size (AUC difference +0.033 [-0.260, +0.326], $p = 0.818$) and, if anything, slightly below homeostasis alone.

[OURS] Unstabilised STDP hurts. In an earlier DoorKey-5x5 experiment, reward-modulated STDP with $\alpha = 1$ solved 0/3 runs against 3/3 for backprop (AUC difference -0.502 [-0.834, -0.170], $p = 0.015$), with a

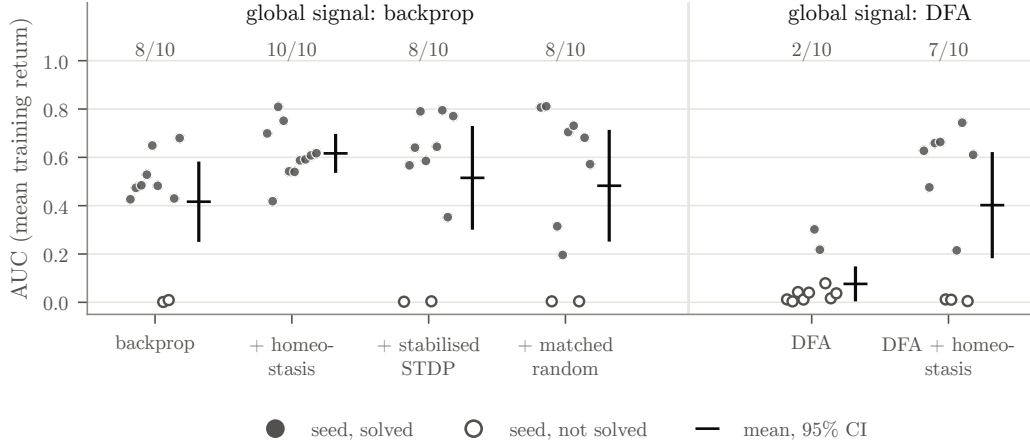


Figure 2: Local-plasticity arms on DoorKey-6x6 (pre-registered; 400k frames, 10 seeds per arm). Each dot is one seed’s AUC (mean training return over the run); filled dots solved the task (final evaluation return above 0.5), open dots did not. Black bars: mean with 95% CI. Numbers above the dots: seeds solved. Left group: backpropagation as the global learning signal, alone or with homeostasis, stabilised reward-modulated STDP, or a random update of the same size. Right group: direct feedback alignment (DFA) with and without homeostasis.

mean spike rate of 0.39 over the last updates against 0.20 for backprop. A larger stabilised α (0.3) was also worse (2/3 solved, context arm). [INTERP] The frame is static across the SNN’s timesteps, so the same pre-post pairings repeat and a mostly potentiating rule feeds back on itself (weights up, rates up, more correlated spikes). Stabilisation removes the runaway but leaves nothing measurable beyond a random perturbation.

[OURS] Inside sleep, the same pattern holds on the SNN trunk (fetch3): vanilla STDP during sleep collapsed performance (ACC 0.163, 1 seed), while stabilised STDP (0.881 ± 0.028) was indistinguishable from plain sleep (0.889 ± 0.003).

5.3 Homeostasis and DFA: faster learning without backpropagation, not reliable learning

[OURS] With backprop, homeostasis raised the AUC by $+0.200$ [$+0.024$, $+0.377$] (raw $p = 0.029$), which does not survive Holm correction ($p = 0.117$); its solve rate was 10/10 vs 8/10 ($p = 0.474$).

[OURS] With DFA as the global signal (pre-registered, 10 seeds per arm), homeostasis raised the solve rate from 2/10 to 7/10 (difference $+0.50$ [$+0.07$, $+0.74$]; Fisher $p = 0.070$, Holm 0.070, not significant) and the AUC from 0.076 ± 0.101 to 0.402 ± 0.307 (difference $+0.326$ [$+0.101$, $+0.551$], 5.3 $\times$; Welch $p = 0.009$, Holm 0.017, significant). The remaining homeostasis seeds still failed outright (Figure 2). [INTERP] Homeostasis is a real but partial fix for DFA: it speeds up learning without a weight-transport requirement, but it does not make DFA reliable on this task, and backprop with homeostasis (10/10) remains better. In continual learning, a DFA+homeostasis encoder with sleep reached only 0.450 ± 0.026 ACC on fetch3.

5.4 Sleep-like consolidation: equal to replay at an equal replay budget

[OURS] Sleep and replay both remove most of naive fine-tuning’s forgetting on every agent (Figure 3, Table 3); EWC protects old tasks at a plasticity cost that grows with λ (return on the last task right after training it: 0.603, 0.441 and 0.280 for $\lambda = 10, 100, 1000$, against 0.971 for naive; CNN). Sleep never differs significantly from replay (last row of Table 3).

[OURS] **Pre-registered, replay budget matched.** On the SNN+Transformer agent, where the original sleep runs had consumed 28% more replayed states than replay, sleep with an exactly matched budget ties replay on accuracy (0.879 ± 0.043 vs 0.862 ± 0.038 ; difference $+0.017$ [-0.026 , $+0.060$], $p = 0.415$, 8 seeds

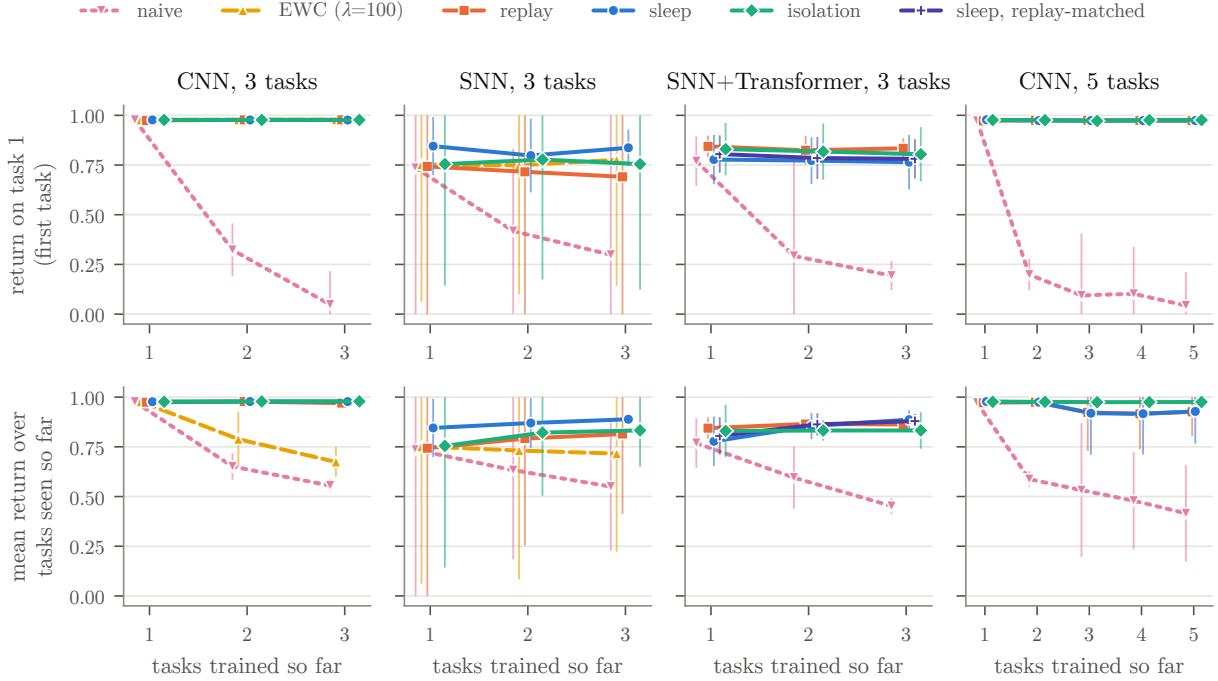


Figure 3: Forgetting on the continual task sequences, by method. Columns: agent and sequence. Top row: return on the first task after training on each task in turn (a flat line means task 1 is not forgotten). Bottom row: mean return over the tasks seen so far. Mean over seeds with 95% CIs (3 seeds per arm, except the SNN+Transformer agent: replay 8, replay-matched sleep 8, sleep 6); the CIs are clipped to $[0, 1]$, the range of returns, and with 3 seeds the SNN intervals span most of it. Methods are offset horizontally so that overlapping lines stay visible. Colours mean the same method in every figure.

Table 3: Final ACC (mean $\pm$ SD over seeds) and FORGET on the continual sequences. “Sleep, matched” consumes exactly as many replayed states as the same-seed replay run (SNN+Transformer only).

	CNN, 3 tasks	SNN, 3 tasks	SNN+Transformer, 3 tasks	CNN, 5 tasks
<i>Final ACC</i>				
naive	0.555 ± 0.013	0.550 ± 0.130	0.452 ± 0.016	0.416 ± 0.098
EWC ($\lambda=100$)	0.674 ± 0.030	0.716 ± 0.199	—	—
replay	0.970 ± 0.009	0.814 ± 0.162	0.862 ± 0.038	0.926 ± 0.049
sleep	0.978 ± 0.001	0.889 ± 0.003	0.887 ± 0.045	0.928 ± 0.065
sleep, matched	—	—	0.879 ± 0.043	—
isolation	0.979 ± 0.001	0.833 ± 0.073	0.832 ± 0.037	0.976 ± 0.000
<i>FORGET (mean)</i>				
naive	0.631	0.321	0.536	0.612
replay	0.002	0.023	0.019	0.006
sleep	0.001	0.006	0.006	0.003
sleep, matched	—	—	0.020	—
<i>sleep – replay, final ACC [95% Welch CI] (unmatched budget)</i>				
	+0.008 $[-0.015, +0.030]$	+0.074 $[-0.328, +0.476]$	+0.024 $[-0.026, +0.075]$	+0.002 $[-0.133, +0.137]$

per arm) and on forgetting (0.020 ± 0.011 vs 0.019 ± 0.017 ; difference $+0.001 [-0.015, +0.017]$, $p = 0.881$) (Figure 4). On the same seeds, the unmatched sleep runs forgot 0.006 on average and the matched ones 0.022; matched minus unmatched ACC was $-0.014 [-0.029, +0.002]$ (paired, descriptive, $p = 0.069$). [INTERP] Sleep’s small earlier edge on this agent came from its extra replay, not from consolidating offline. This matches

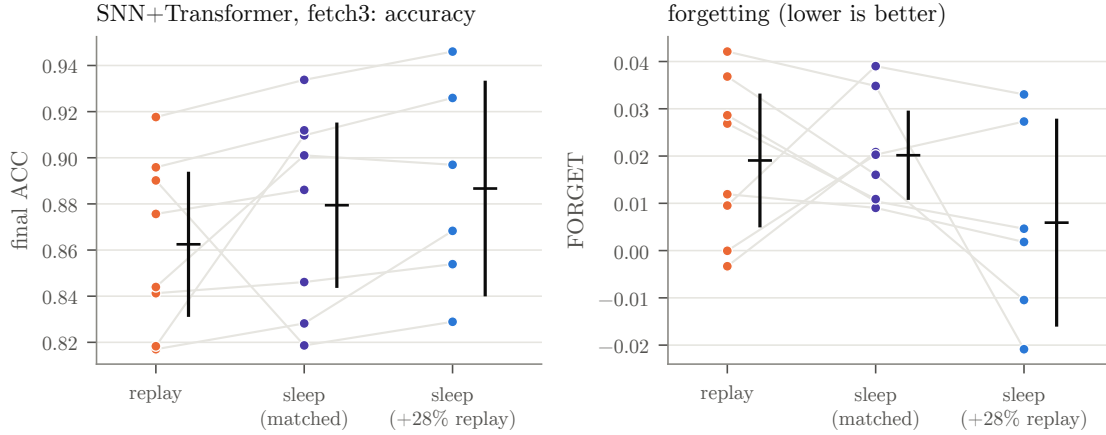


Figure 4: Sleep versus replay at an exactly matched replay budget (SNN+Transformer agent, fetch3; pre-registered, 8 seeds per arm). *Replay*: CLEAR-style replay. *Sleep (matched)*: each run replays exactly as many stored states as the replay run with the same seed. *Sleep (+28%)*: the original sleep runs, which replayed 28% more states. Grey lines connect runs with the same seed; black bars are means with 95% CIs. Left: final accuracy (ACC). Right: forgetting (FORGET; lower is better).

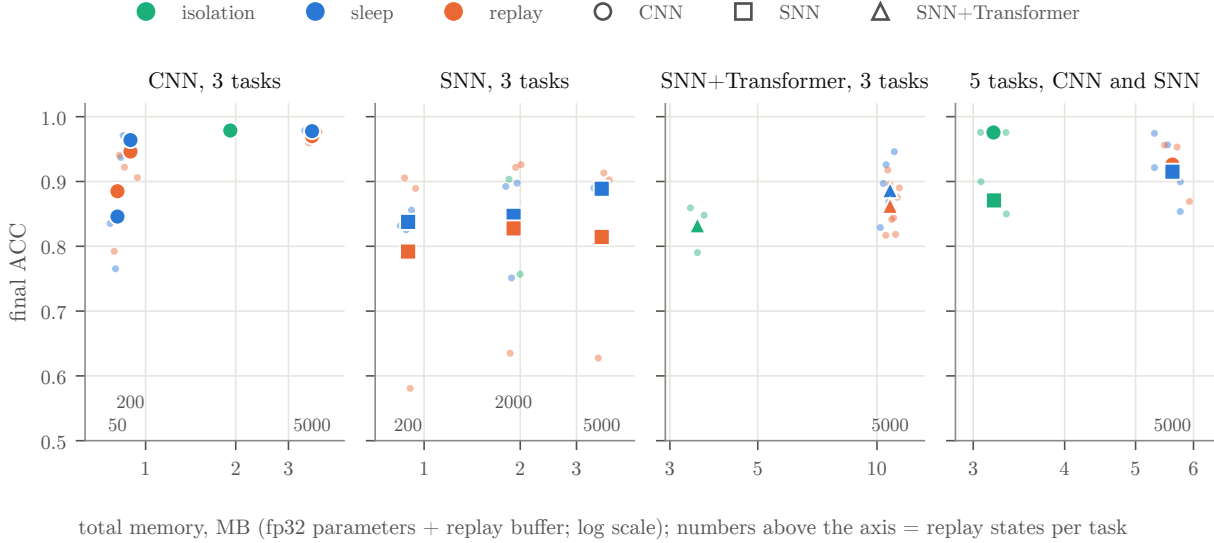


Figure 5: Final accuracy (ACC) against total memory: fp32 parameters plus the replay buffer at its measured storage cost, log scale. Small dots: seeds; large markers: means. Numbers above the x -axis: replay states stored per task by sleep and replay; isolation stores none but trains one network per task. At the default 5,000 states per task, isolation needs the least memory; with 200 states per task, sleep reaches a similar mean accuracy in less memory (3 seeds; wide CI). The 5-task SNN runs use the fair budget of 450k frames per task (Section 6).

the literature point that replay is a very strong baseline (Rolnick et al., 2019): in our implementation sleep is a way of scheduling replay, and at equal replay it is not better.

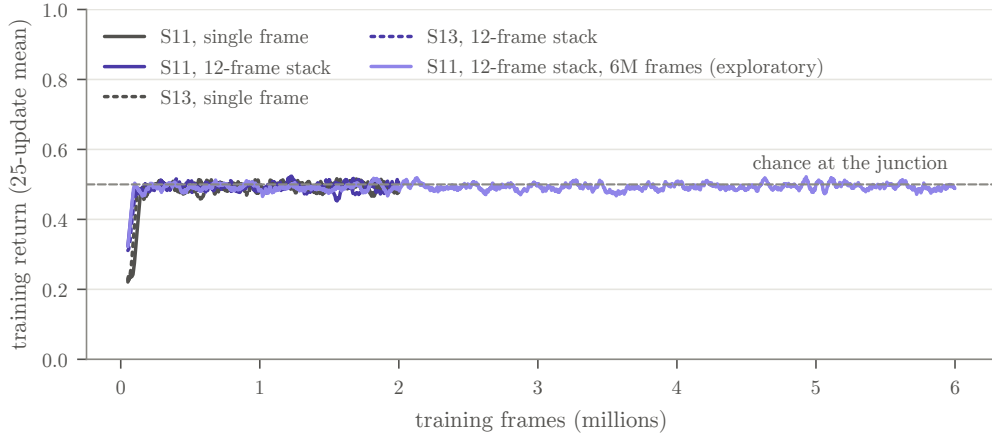


Figure 6: Memory-benchmark probes (pre-registered sizing runs; CNN encoder, one seed each, learning rate 0.0003). Training return (moving mean over 25 updates) on MiniGrid-MemoryS11 and S13 with a single frame (grey) or a stack of the last 12 frames (violet); light violet: an exploratory 6M-frame run of the stacked CNN on S11. A return of 0.5 is chance at the junction. No run learns to use the cue, so neither map tests working memory.

5.5 Shared weights versus isolation: memory-efficient only with small buffers

[OURS] Per parameter, the shared agent with sleep is clearly more efficient than isolation: ACC per 100k parameters 0.50 vs 0.20 (CNN), 0.46 vs 0.17 (SNN), 0.27 vs 0.09 (SNN+Transformer), 0.41 vs 0.12 (CNN, 5 tasks). But at the default 5,000 stored states per task the buffer outweighs the extra networks: isolation uses less total memory and gets more accuracy per MB everywhere (0.51 vs 0.27, 0.43 vs 0.25, 0.24 vs 0.08, 0.31 vs 0.17 per MB; Figure 5).

[OURS] With small buffers the shared agent can be the more memory-efficient one. On the CNN, sleep with 200 states per task reaches 0.964 ± 0.009 in 0.89 MB (1.08 per MB, against 0.51 for isolation). On the SNN, sleep with 200 states per task reaches a similar mean accuracy to isolation (0.838 ± 0.016 vs 0.833 ± 0.073 ; difference $+0.005$ [-0.167 , $+0.177$], $p = 0.921$, 3 seeds per arm) in 0.89 MB against 1.92 MB, 46% of the memory. With 3 seeds the interval is wide: the data rule out neither a large deficit nor a large advantage, so this is not evidence of equal accuracy. At *equal* total memory (SNN, 2000 states per task, 1.91 vs 1.92 MB) sleep does not beat isolation (difference $+0.014$ [-0.165 , $+0.193$], $p = 0.836$).

[OURS] Forward transfer depends on the encoder: positive where learning from scratch is slow (SNN 0.199 ± 0.079 , $p = 0.049$; SNN+Transformer 0.423 ± 0.244 , $p = 0.095$; 5-task SNN 0.306 ± 0.170 , $p = 0.089$), near zero on the 3-task CNN (-0.017 ± 0.141) and negative on the 5-task CNN (-0.525 ± 0.722). [INTERP] The shared agents’ advantage is memory efficiency when replay is cheap, not higher accuracy at equal memory. “Shared weights are more efficient than isolation” is only true with the buffer size stated.

5.6 Transformer working memory: untested, because the benchmark failed

[OURS] On the fetch tasks, which are fully determined by the current frame, the Transformer has nothing to remember; the SNN+Transformer agent needed 400k frames per task where the CNN and SNN agents used 150k. To test memory we used MiniGrid-Memory. On S7 (pilot, one seed, 1M frames) a memoryless CNN already succeeded in 0.915 of episodes (a 4-frame stack: 0.995), so S7 does not isolate memory: in that small map the cue and both choices are visible from one position. The pilot’s SNN runs (SNN with an 8-frame stack 0.465, SNN+Transformer 0.970) used different learning rates (0.001 vs 0.0003) and cannot support a claim.

Table 4: Earlier positive findings that shrank or disappeared once a control was added. *Evidence then*: the comparison as it stood, recomputed from the frozen runs restricted to the seeds available at the time. *After the control*: the same comparison with the control in place. *p*: Welch unless noted.

Earlier finding	Evidence then	After the control	Cause
Sleep beats replay (SNN+Transformer)	seeds 1-3: +0.039 [-0.006, +0.084], $p = 0.072$	6 seeds: $p = 0.224$; budget-matched, 8 seeds: +0.017 [-0.026, +0.060], $p = 0.415$	seed count; replay-sample budget
Shared weights far more efficient than isolation	ACC per parameter: 2.5-3.3×	per total memory at the default buffer, isolation is 1.8-2.9× more efficient; with 200 states/task (SNN), similar mean ACC (0.838 vs 0.833; 3 seeds, wide CI) in less than half the memory	memory accounting
Sleep far beats isolation on 5 tasks (SNN)	150k/task, 2 seeds: 0.778 ± 0.007 vs 0.475 ± 0.023, $p = 0.023$	450k/task, 3 seeds: 0.915 ± 0.014 vs 0.871 ± 0.026, +0.044 [-0.009, +0.098], $p = 0.078$	training budget
Homeostasis helps backprop	3 vs 5 seeds (homeostasis vs backprop): 3/3 solved, $p = 0.088$	10 seeds: +0.200 [+0.024, +0.377], $p = 0.029$, Holm 0.117	seed count; multiple comparisons
Homeostasis makes DFA work	3/3 vs 1/5 solved, AUC $p < 0.001$	10 seeds: 7/10 vs 2/10, Fisher $p = 0.070$ (AUC effect survives)	seed count
Transformer helps on a memory task	S7 pilot: 0.970 vs 0.465 (frame stack)	memoryless CNN on S7: 0.915; on S11/S13 no memory agent learns	benchmark validity; learning-rate confound

[OURS] For the pre-registered benchmark we probed S11 and S13, where the cue is further from the junction than the view reaches. The memoryless CNN was at chance, as required (0.430 and 0.485 success), but so was a CNN with a 12-frame stack (0.465 and 0.460), with episodes of about 10-12 steps (Figure 6). Neither map met the pre-registered criterion, so the Transformer comparison was not run. An exploratory run (not pre-registered) of the stacked CNN on S11 for 6M frames stayed at chance (0.485; training return 0.48-0.51 throughout). [INTERP] The episode lengths show that no agent learned to go back and look at the cue; they walk to the junction and guess. The obstacle is exploration and credit assignment, not context length. We discuss this in Section 7.

6 What did not survive controls

Several results looked positive before the controls were in place. Table 4 lists them with the control that removed them. All “before” numbers are recomputed from the same frozen runs, restricted to the seeds that existed at the time.

Seed count. [INTERP] A handful of seeds was enough to produce $p = 0.072$ (sleep vs replay) and $p = 0.088$ (homeostasis vs backprop), and a 3/3 vs 1/5 split that looked decisive (DFA). All three shrank once the seed count was raised. With bimodal outcomes, one failed seed moves a small-sample mean a lot.

Training budget. [INTERP] At 150k frames per task, a freshly initialised SNN often had not learned its task yet, so isolation (which starts from scratch on every task) was under-trained while the shared agent benefited from transfer. At a budget where fresh networks learn (450k, chosen by single-task probes before the runs), the gap shrank from +0.303 [+0.148, +0.457] to +0.044 [-0.009, +0.098]. Sleep still learned more tasks to 0.8 or above (14/15 vs 12/15), which is consistent with some positive transfer.

Memory accounting. [INTERP] Comparing per parameter ignores the replay buffer, which is the price the shared methods pay for not forgetting. With the buffer counted, the conclusion reversed at the default buffer size.

Replay-sample budget. [INTERP] Sleep and replay are usually compared at equal buffer size, but the number of replayed samples they consume can differ a lot (here by 28% on average). Matching it removed the difference.

Benchmark validity. [INTERP] A memory benchmark must fail without memory and be learnable with it. S7 failed the first condition; S11 and S13 failed the second at our budget.

7 Limitations

Task identity is given. Our setting is task-incremental: the agent knows which task it is doing. Removing the task identity from fetch3 would make it unsolvable (the goal is not visible), so a task-agnostic study needs a different benchmark, and sleep’s distillation, which relies on per-task heads, would need redesigning.

Scale. All agents are small (159,864–327,800 parameters), the environments are MiniGrid gridworlds, sequences have 3 or 5 tasks, and budgets are 150k–450k frames per task. Results may not transfer to larger agents, longer sequences or visually rich tasks.

CPU only, few seeds. Everything ran on a 4-core CPU. Many continual arms have only 3 seeds, so their CIs are wide and ties mean “no large difference”, not equivalence. The pre-registered comparisons used 8–10 seeds.

Energy is a proxy. The energy numbers multiply operation counts by literature constants for 45nm arithmetic and assume event-driven hardware. We measured no power. On the CPU we used, the SNN was slower.

Single settings. Hyperparameters (homeostasis target, α , EWC λ beyond three values, sleep schedule) were not tuned per arm. A different homeostasis target or sleep schedule could change the results.

Baselines are simple. Isolation uses one full network per task, not PackNet-style masks, which would be cheaper and a stronger baseline. Our sleep phase is replay plus distillation, not the sleep replay consolidation of Tadros et al. (2022).

The Transformer question is untested. On S7 memory was unnecessary; on S11 and S13 no agent, including a CNN with a 12-frame stack, learned to use the cue, even at 6M frames. A usable benchmark needs the cue to be seen on the way to the decision (e.g. starting in the cue room facing the corridor), an exploration aid, or a much larger budget. The Transformer comparison, and the GRU baseline we implemented for it, remain future work.

Pre-registration was informal. The pre-registered plans were recorded in the project’s version-controlled log before the runs, not in a public registry. The first two phases were exploratory, and their positive findings are exactly the ones Section 6 revisits.

8 Conclusion

We isolated seven brain-inspired mechanisms in a small continual RL setting and gave each a matched control. [OURS] None improved final accuracy or forgetting beyond its control: STDP was no better than a random update of the same size, and sleep was no better than replay at an equal replay budget. There were two specific exceptions. Homeostasis sped up learning of a spiking encoder trained by DFA instead of backpropagation, without making it reliable, and with a small replay buffer a shared network with sleep

reached similar mean accuracy to isolation (0.838 vs 0.833; 3 seeds, wide CI) in less than half the memory, while at the default buffer size isolation was more memory-efficient. The spiking encoder offered access to a lower operation-count regime, whose energy value depends on hardware we did not test, and the Transformer question remains open. [INTERP] The most consistent lesson is methodological: in this setting, the earlier positive effects of biological mechanisms shrank or disappeared once seed counts, training budgets, replay budgets and memory accounting were made fair; the speed-up that homeostasis gives DFA survived. We suggest that claims of this kind are reported with those budgets stated and matched.

Reproducibility statement

All code, configuration files, the frozen run directories and the analysis scripts are in an anonymised repository provided as supplementary material. For each of the 289 PPO runs the repository keeps the full configuration, the training log, the evaluation results and the final checkpoint, so every number can be traced to its run and every run repeated with its seed. There are three levels of reproduction.

1. *The paper from the frozen results* (minutes, no training). `python paper/compute_numbers.py` regenerates every number in the text; `paper/NUMBERS.md` lists each one with its source file. `python paper/figures/make_figures.py` regenerates the figures, and `latexmk -pdf main.tex` builds the paper. The statistical tests are rerun by `scripts/rq1_stats.py`, `scripts/rq5_stats.py`, `scripts/s4_stats.py` and `scripts/memory_fair.py`.
2. *A single run.* `python train.py --config configs/<name>.yaml --set seed=<s>` for a single task and `python continual.py --suite fetch3 --method <method> --config configs/<name>.yaml --seed <s>` for a task sequence; any stored configuration key can be overridden with `--set`.
3. *All experiments.* The job lists in `jobs/` run through `scripts/run_queue.py` (resumable, one thread per job). The frozen runs took about 110 single-thread CPU hours in total, on a 4-core machine without a GPU.

The software environment is Python with PyTorch (CPU), Gymnasium and MiniGrid (`requirements.txt`); the unit tests run with `pytest tests/`. The `demo/` folder holds an offline results page, GIFs of the naive and sleep agents after the full fetch3 sequence (final accuracy-matrix rows 0.126, 0.542, 0.956 and 0.978, 0.979, 0.979), and a script that runs a trained agent for one episode on CPU.

AI assistance disclosure

TODO: Author to write: which parts of the code, experiments, analysis and text were produced with AI assistance, and how they were checked.

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A Protocol constants

PPO	16 envs $\times$ 128 steps, 4 epochs, minibatch 256, clip 0.2, $\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$, KL stop 0.03
Learning rate	0.001 (CNN, SNN), 0.0003 (SNN+Transformer, memory probes)
Frames per task	150k (CNN, SNN), 400k (SNN+Transformer), 450k (5-task SNN, fair)
SNN	4 timesteps, learnable leak (init 0.9), threshold 1, surrogate slope 25, membrane reset per frame
Plasticity	$\alpha = 0.03$ (pre-registered), homeostasis target 0.15, rate 0.01
Sleep	every 25 updates + end of task, 700 steps per phase, 5,000 states per task
Evaluation	100 episodes per task (continual), 200 (single task), sampled actions
Memory accounting	188 B/state (CNN, SNN), 632 B/state (SNN+Transformer)

B Additional numbers

EWC on the CNN (3 tasks): ACC 0.816 ± 0.120 , 0.674 ± 0.030 and 0.545 ± 0.006 for $\lambda = 10, 100, 1000$ (forgetting -0.005 , -0.002 , 0.000). Isolation on 5 tasks at the fair budget: 3.20 MB, ACC 0.871; sleep 5.62 MB, ACC 0.915. SNN replay with small buffers: 200 states per task 0.792 ± 0.183 , 2000 states per task 0.828 ± 0.167 . The complete list of numbers, with sources, is `paper/NUMBERS.md`.